# David Jovany Marin Holguin

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## EDUCATION

**May 2027**

**University of Kentucky** **Lexington, KY**

*Bachelor of Science in Biomedical Engineering* Major GPA: 3.6

## INTERNSHIP EXPERIENCE

**Johnson & Johnson Ethicon** **Cincinnati, Ohio**

*Design Engineering Co-op* Spring 2026- Present

- Supported lifecycle management for 10 harmonic and advanced bipolar surgical devices by applying design controls and developing SolidWorks models, ensuring regulatory compliance and reducing design change requests across 15+ projects.
- Led verification testing, updated test methods, and performed statistical analysis across eight product families, supporting data-driven decisions that improved product reliability and manufacturability.
- Collaborated cross-functionally with manufacturing, quality, design, and cost-savings teams to minimize production line downtime and improve operational efficiency
- Contributed to engineering documentation and standard operating procedures, supporting continuous improvement initiatives that streamlined laboratory safety compliance.

**Atlantis: Clinical Immersion** **Rome, Italy**

*Clinical Shadowing- Orthopedic Rotation* Spring 2025

- Completed 100+ clinical shadowing hours observing medical procedures and device usage, strengthening understanding of clinical applications and patient outcomes.
- Gained exposure to sterile environments and procedural workflows.

**Shriners Medical Hospital** **Lexington, Kentucky**

*PT/OT Intern* & Prosthetics Spring 2023 & Fall 2025

- Implemented evidence-based therapy protocols and monitored patient progress through motion analysis, contributing to enhanced treatment outcomes for 12+ patients.
- Assisted in design verification and validation by seeing adjustments of the prosthesis and choice of biomaterials to improve mobility and long-term quality of life, reinforcing risk management and design verification.

## WORK EXPERIENCE

**All My Sons Moving** **Lexington, Kentucky**

*Driver/On-site Manager* June 2023- Present

- Optimize teamwork and load organization to ensure secure transport, reducing customer claims. Leading branch customer satisfaction reports within the branch for 2+ years.
- Leading region in safety initiatives to improve driver metrics by 5% and increase overall branch revenue by 3%, led by applying clear safety regulations and keeping a clean working environment.

## UNIVERSITY PROJECTS

**Lexington, Kentucky**

**Senior Design- Adaptive Pressure Distribution Orthotic Insert** Fall 2026

- Developing a low-profile orthotic platform to address abnormal plantar loading through patient-focused pressure redistribution, structural support, and localized sensing concepts.
- Translate sponsor, clinician, and stakeholder needs into engineering requirements, acceptance criteria, material selections, and design concepts for an orthotic medical device.
- Evaluate a unidirectional anisotropic carbon-fiber structural layer, pressure-relieving materials, and replaceable sensing patches while considering biomechanics, manufacturability, usability, and verification strategy.

**Design & Development of Knee Orthotic (Wearable Medical Device)** Fall 2025

- Collaborated to design and develop an innovative knee orthotic for athletes, integrating electro stimulating muscle therapy for passive recovery.
- Integrated design thinking to conduct prototype testing and evaluation, analyzing performance and clinical usability for athlete recovery.
- Performed CAD modeling in SolidWorks, material choice, and multiple design iteration with user feedback to optimize functionality and manufacturability.

**ECG Data Analysis & Respiratory Rate Detection** Fall 2025

- Manufactured MATLAB algorithms to receive raw ECG data and calculate user respiratory derived from R-peak detection using engineering analysis, validation, and documented results in a formal technical report.

## CERTIFICATIONS AND TECHNICAL SKILLS

- True LEAN by Toyota: University of Kentucky
- Data Analysis: Minitab, MATLAB, Excel, Statistical Analysis
- Documentation: Laboratory notebooks, test protocols, validation reports
- Quality Systems: Understanding of regulated environments, validation processes, and documentation control
- CAD & Simulation: SolidWorks, Siemens NX, Onshape
- NOCTI Lab certified (Laboratory safety, Equipment Handling)
- Harvard Medical Pro Certification of Achievement: Pharmacology Drug Delivery
- AAPPL Seal of Biliteracy: Spanish & English